Adam Boyher

Contact

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Phone: (636)-236-8870

Location: St. Louis Mo, USA

SKILLS

Python, R, Unix, C++, Java, Javascript, Julia, SQLite, AWS (Lambda, ECR, ECS, Batch, Step Functions, EventBridge, Load Balancing), Elasticsearch, Sklearn, TensorFlow, PyTorch, Machine Learning (SVM, BERT, KMM), Plotly Dash, Docker, MKDocs, Nextflow, Anaconda, Django, RShiny, HTCondor, Snakemake, BUSCO, Mercury, DESeq2, Falcon suite, Juicebox, 3d-DNA, Maker Annotation, Domino Data Lab

EXPERIENCE

Bayer Crop Science — *Computational Scientist*

December 2020 - PRESENT

- Core member of the Science, Competitor, and IP Intelligence (SCIPI) team of IPB, contributing to digital solutions for large-scale text data processing and visualization.
- Designed and implemented scalable infrastructure with AWS services (Lambda, ECS, Step Functions, Batch, EventBridge), Docker, and load balancing for seamless integration and deployment.
- Developed the Crop Science Publication Dashboard, a live dashboard for exploring publications by Bayer employees.
- Built machine learning models (using Sklearn SVMs and BERT) with Django, Plotly and Streamlit frameworks for advanced text classification to monitor project specific publications for Regulatory Science.
- Developed tool called Expert Finder for ranking scientific authors and institutions within specific research domains based on publication data.
- Developed Python package to streamline access to Crossref, Altmetric, OpenAlex, and various internal tools for data analysis.
- Led the development of a team documentation site for computational resources and pipelines using MKDocs, enhancing access to project documentation and technical guides.
- Mentored a summer intern in 2021, guiding them in data science workflows and best practices in computational science.
- Active participant in BayHack 2022, an internal Bayer hackathon.

Donald Danforth Plant Science Center — *Senior Computational Scientist*

March 2018 - December 2020

- De novo genome assembly
- Writing Nextflow pipelines for GATK, genome assembly, annotation
- AWS cloud computing
- Development of apps for viewing RNAseq and genomic data
- Perform differential expression analysis on RNAseq data
- Developed pipeline for performing Dual RNAseq analysis

University of Washington, St. Louis — *Data Coordinator*

January 2018 - March 2018

- Coordinated data collection with RedCap for the Orthopedic Surgery Department

Donald Danforth Plant Science Center— *Greenhouse Technician II*

October 2015 - December 2017

- Built and maintained the Plant Growth Facility website
- Managed automated watering and environment control systems
- Managed 9 research greenhouses for internal and external clients general plant care, insect and disease control

Donald Danforth Plant Science Center — *Laboratory Technician I*

January 2014 - October 2015

- Designed and conducted greenhouse experiments
- Developed a protocol for testing herbicide tolerance of cassava (*Manihot esculenta)* and scoring of disease symptoms.

EDUCATION

Georgia Institute of Technology — *M.A. Computer Science*

Current

University of Missouri, Columbia — *B.A. Biology*

2011 - 2012

University of Missouri, Columbia — *B.A. Mathematics*

2006 - 2010